

Ashwin Deshpande

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Google Scholar: <https://scholar.google.com/citations?user=b6j5q5sAAAAJ&hl=en&authuser=1>

SKILLS

Languages: Python, R, SQL, Shell, C

Data Science: FinTech, A/B testing, Machine Learning, Statistics, Probability, NLP, Recommendation Systems

ML/AI: Scikit-learn, TensorFlow, Keras, PyTorch, Scipy, SHAP, BERT, LLMs, Hyperparameter Tuning, Feature Engineering, Time Series Modeling

Data Management: PySpark, Mlflow, Pandas, Numpy, BigQuery, PostgreSQL

Databases & Tools: Snowflake, PostgreSQL, Athena, DynamoDB, BigQuery, MySQL, Redis

Data Visualization: Looker, PowerBI, Matplotlib, Seaborn, ggplot2, Bokeh, Plotly, Shiny

EXPERIENCE

Data Scientist

Thynk Capital LLC, San Jose, CA | May 2023 – Feb 2025

- Architected a fully automated hedge fund **analytics** and trading platform utilizing distributed **data pipelines** and containerized **ML** microservices in a hybrid cloud environment (**AWS/GCP**) and **Interactive Brokers API**.
- Engineered 20+ **quantitative** trading features such as **Bollinger Bands**, **RSI**, **MACD**, and custom **greeks** using **PySpark**, streamlining feature engineering for model experimentation
- Developed comprehensive **feature** set to make informed trade decisions based on S&P Index & its sectors, NASDAQ, DJIA, US Treasury, Russell, Gold, Oil, VIX & Exchange rates
- Developed multiple interdependent **multi-class classification** sequential models (**Transformers**) predicting **delta**, **gamma** & implied volatility(**VIX**) for daily **option** trades
- Designed high quality gold target data based on historical performance for predicting SPY ETF direction and volatility from next 3 to 30 days.
- Built a high-throughput **simulation** engine for **backtesting** options strategies over 12 years of historical data, using **GCS** checkpoints and **Bayesian optimization** for tuning model hyperparameters efficiently.
- Designed and implemented scalable **ETL** pipelines ingesting **>1TB/day** of minute-resolution trading data from multiple exchanges using **AWS Lambda** and **PySpark** with auto-triggered batch processing via **S3** events.
- Built reusable **ETL** templates using **Apache Airflow** by developing a **Python**-based OOP wrapper framework that parameterized **DAGs** through **YAML** configurations, improving development agility and reducing DAG boilerplate by 80%.
- Reduced compute costs by 60–70% on **GCP** by optimizing **Dataproc** configurations, deploying autoscaling **Spark** clusters, and tuning executor memory and core usage based on workload size.
- Created unified data lake and data warehouse architecture using **S3**, **GCS**, and **BigQuery**. Optimized analytical performance through dynamic **partitioning**, **clustering** on high-cardinality columns, and query rewrite logic.
- Developed **RESTful** ML model serving APIs for quantitative predictions using **Flask**, **Docker**, and **Kubernetes**. Implemented canary deployments and resource autoscaling in **GKE** clusters to handle real-time inference with minimal latency.
- Spearheaded **CI/CD** integration using **Cloud Build**, **GitHub Actions**, **Artifact Registry**, and automated testing pipelines, ensuring robust and frequent deployments.
- Performed cost-benefit analysis and optimization across multi-cloud platforms to inform financial decisions related to data transfer, storage, and compute resources.

Data Scientist (Internship)

Bombora Inc, Chicago, IL | June 2022 – Dec 2022

- Designed scalable data pipelines using **PySpark** to process behavioral intent data from B2B digital signals for clients such as IBM, Oracle, and Adobe.
- Resolved critical model performance issues by identifying data leakage in **feature generation** pipelines. Developed custom **sklearn**-compatible transformers to apply time-window-based validation.
- Developed a model evaluation dashboard combining univariate analysis, **SHAP** plots, and temporal performance trends using **Jupyter**, **Looker**, and **Matplotlib**.
- Achieved 8% improvement in **Random Forest** precision through hyperparameter tuning and feature selection, resulting in reduced false positives and improved client satisfaction.
- Created **Looker** visualizations to identify high-cost data vendors with minimal contribution to model lift, resulting in substantial reduction in third-party data acquisition costs.

Graduate Research Assistant – NLP Lab

University of Illinois at Chicago | Jan 2021 – May 2022

- Led the development of a speech-based cognitive screening system to identify early-stage dementia based on disfluencies and speech gaps.
- Integrated **IBM Watson** and **Google STT APIs** with custom retraining pipelines to improve transcription accuracy for impaired or disfluent speech.
- Engineered **NLP** pipelines to extract syntactic, acoustic, and semantic features using **spaCy**, **BERT** embeddings, and signal processing techniques.
- Built deep learning classification models using **TensorFlow** and **PyTorch** to detect cognitive decline.
- Achieved publication in **ACL-BioNLP 2022**: <https://aclanthology.org/2022.bionlp-1.4/>
- Contributed to open-source medical **NLP** tools and collaborated with linguists to validate research findings.

Data Scientist

Primenumbers Technologies, Bengaluru, India | May 2019 – May 2020

- Designed and deployed scalable web scraping systems using **Python**, **Selenium**, and **PostgreSQL** to gather and normalize tender documents from global government portals.
- Developed **NER** pipelines with **spaCy** and custom-trained word embeddings to extract key metadata (organizations, dates, monetary values, locations).
- Improved **Solr**-based search engine performance by implementing domain-specific stemming and synonym mapping, resulting in a 5% **CTR** boost via **A/B testing**.
- Conducted regular performance benchmarking and data quality audits, resulting in a 15% improvement in downstream accuracy for internal users and clients.

PROJECTS

UIC Search Engine – GitHub: [\[GIT\] Semantic vs Lexical Matching](#)

- Developed a topic-specific search engine for UIC domain using custom-built **web crawlers** and **inverted index**.
- Designed semantic search using **BERT** embeddings and **lexical** search using **TF-IDF** to provide hybrid relevance scoring.
- Built a scalable **data ingestion pipeline** using multithreaded crawling and scheduled batch jobs to ensure up-to-date **indexing**.
- Implemented **PageRank** and **HITS** algorithms to compute ranking signals based on web link graph and domain authority.
- Utilized **Flask** and **Elasticsearch** for backend query processing and frontend integration.

- Used **Airflow** to orchestrate crawling and indexing jobs and scheduled **Spark** jobs for large-scale document embedding

Customer Churn in FinTech Firm – GitHub: [\[GIT\]](#)

- **Feature engineered** signals to predict if a customer is likely to stop using product based customer clickstream behaviour
- Used **explainable** ML technique to determine causes of customer churn by looking at feature importance
- Developed **polynomial features** to capture non-linear interaction between predictors and target churn variable
- Achieved 0.92 **ROC AUC** in detecting customer churn

Conversational Recommender for Movie Recommendations – GitHub: [\[GIT\]](#)

- Solves cold-start problem in **recommendation systems** by adding a **task-oriented dialogue system** to understand user preferences
- Developed a **Knowledge-graph** based **embeddings** to capture similarity between user's recent dialogue request & historical preferences to a ranked list of movie entities

Spotify ETL Pipeline – GitHub: [\[GIT\]](#)

- Engineered a batch-based **ETL** pipeline to ingest **JSON** data from Spotify **REST API** and transform/clean playlist metadata.
- Employed **AWS Lambda** for compute, **S3** for storage, and **DynamoDB** for **NoSQL** querying with Kimball-style dimensional modeling.
- Implemented **Athena**-based **SQL** access layer on top of **S3** for large-scale data exploration using **AWS Glue** crawlers and partitioning.
- Exposed APIs via **AWS API Gateway** for **RESTful** access; integrated with **CRON** triggers for scheduled updates.
- Added monitoring via **CloudWatch** and cost alerting using **AWS Budgets**.

Real-Time IoT Data Pipeline – GitHub: [\[GIT\]](#)

- Simulated high-throughput sensor data streams (temperature, humidity, pressure) from virtual **IoT** devices on **AWS EC2**.
- Built a real-time ingestion pipeline using **Apache Kafka** as message broker and **Spark Streaming** for micro-batch processing.
- Configured **Kafka** clusters with **Kubernetes HPA** for elastic scaling based on incoming message volume.
- Integrated **Amazon Kinesis** as a buffering layer to manage bursty **Kafka** workloads.
- Stored processed data into **Snowflake** and **PostgreSQL** for downstream analytics and visualization.
- Visualized real-time metrics using **Grafana** dashboards connected to **PostgreSQL**.

EDUCATION

- **University of Illinois, Chicago**
 - GPA: 4.0 - *M.S. in Computer Science* - Aug 2020 - May 2022
- **B.V.B College of Engineering & Technology, Hubli**
 - GPA: 3.5 - *B.Eng in Computer Science* - Aug 2014 - May 2018

CERTIFICATIONS

[Linear Algebra](#)
[Bayesian Statistics](#)
[Deep Learning.ai-Intro to MLOps](#)
[Deeplearning.ai-ML Data Lifecycle](#)